Visual Data Predictions with Multimodal Topic Mining

Student Details

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Dataset: Yu-Gi-Oh! Dataset YGOProDeck API

Colab Implementation

Google Colab Link:

https://colab.research.google.com/drive/1YsEDRujNl0jBQNz1xcBEZBueEZE_s7lB?usp=sharing

This assignment implements multimodal topic mining using BERTopic with CLIP embeddings on Yu-Gi-Oh! trading cards dataset. The notebook includes:

1. Data Collection: Downloaded 8,000 Yu-Gi-Oh! cards from YGOProDeck API with images and metadata.
2. Topic Mining: Implemented topic_miner() function using BERTopic with CLIP (clip-ViT-B-32) for visual topic discovery.
3. Prediction: Implemented predict() function with Random Forest classifier using 70-30 train/test split.
4. Comparison: Evaluated three approaches - image-only features, card game features (ATK, DEF, race, type), and combined multimodal features.